Transfer task: defend and revise

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Support your prediction with measurements and calculations. Show a specific change and explain why you made it.

SUCCESS CRITERIA - I can...

1. Graph the central prediction and coordinate bounds.
2. Use calculations to support an answer. Explain when the calculation stops working.
3. Show a specific change and use measurements or details to explain why you made it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your answer. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

A student says a **3 kg** cart under **15 N** net force accelerates at **45 m/s²**. **Name the error**, correct it and **name a check you can** now do independently.

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Use your Day **21** data and consistent units. **Explain the endpoint** and what you treat as true. A bounded range is not a probability.

Worked reasoning: The cart calculation predicts **12 m** after **4 s**. The measured position is **10 ± 0.5 m**. Since **12 m** is outside **9.5–10.5 m**, check the calculation and what it treats as true. The difference alone does not tell you its cause.

Your turn: Mark the observed interval and prediction on a number line. **Name one thing the calculation** treats as true to recheck.

Sentence starter: I predict … . I used … to make this prediction because … .



Position (m); right is positive

Marzano check - circle: ① ② ③

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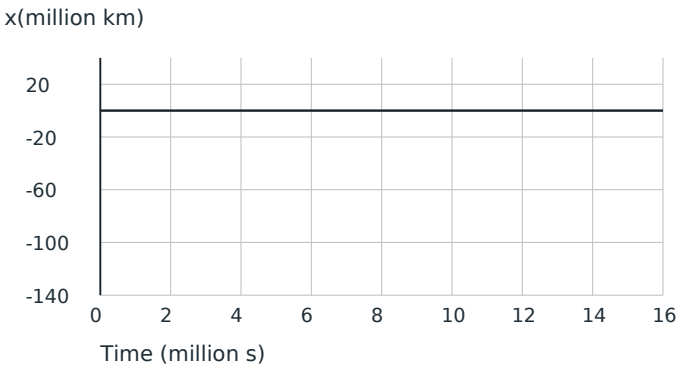
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Show the prediction

Using your Day **21** CPA work, draw an x-versus-time graph. Connect the central start and finish. Then **add lower** and upper prediction lines. **Label time** in millions of seconds and x in millions of km.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③

04 Write a public briefing

Write **4–5 sentences**. Include the central predicted position, the range of possible positions, one thing the prediction treats as true, and one physical effect it leaves out. **Explain what you can** and cannot conclude about an Earth impact.

Sentence starter: I think … . I can point to …, which shows …, so … .

Explain to your audience	Evidence from your work
Predicted position	____
Range and meaning	____
Assumption / omitted effect	____

Marzano check - circle: ① ② ③

05 Defend and revise

Ask a partner to **check one calculation** and one graph label. **Record one question**, then revise a specific step or reason why it should stay. If no partner is available, perform and **label a self-review**.

Support: I changed (or kept)____ because ____.

Sentence starter: My design includes … . It does what is needed by … .

Step checked	Question or concern	Change or reason to keep
____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A **6 kg** sign hangs still from one vertical cord. Use g=**10 N/kg**. **Name both forces**, their sources, magnitudes and directions.
Draw the forces on the chosen object and **label their sources**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

1. Point to one piece of your work that meets each success criterion.
2. **State one conclusion you can** now support.
3. **Name one question about curving** paths that needs ideas from Unit **2**.

Sentence starter: I notice ... . This supports ... because ... .

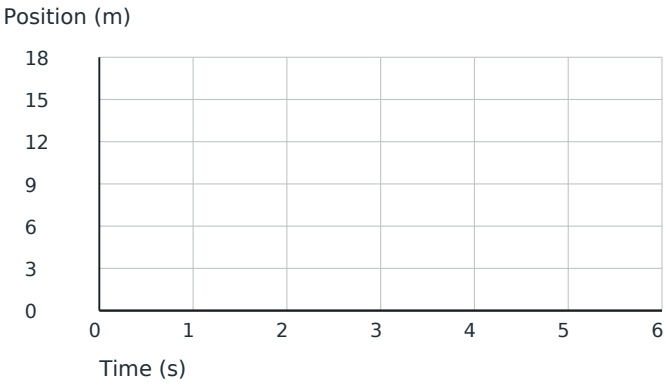
Criterion	Where your work shows it
Science / model	____
Reasoning	____
Communication / transfer	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Revise a constant-speed calculation from a **0 m** start to match the central observation of **10 m** after **4 s**.
2. Propose a new observation that could test the revision.

Sentence starter: My design includes ... . It does what is needed by ... .



Marzano check - circle: ① ② ③